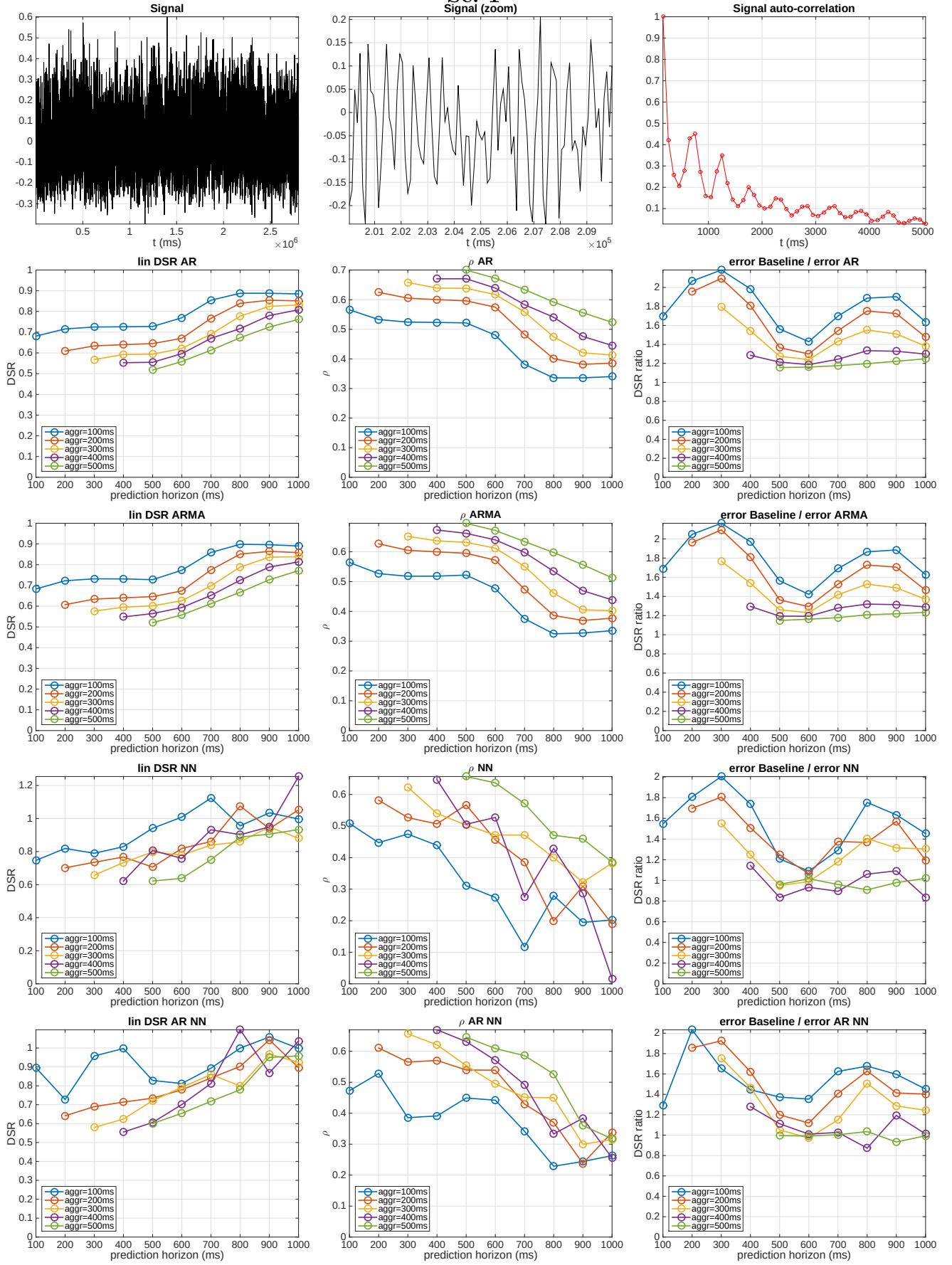
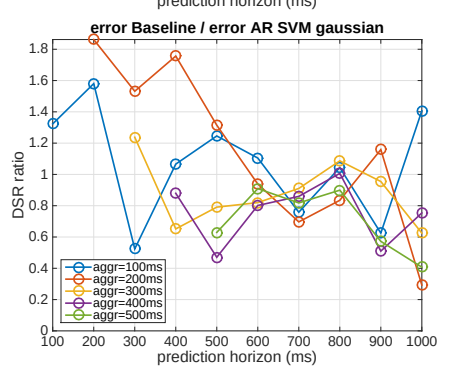
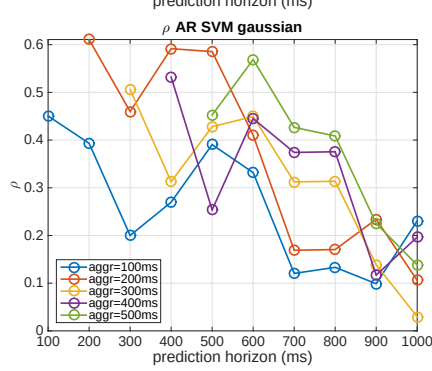
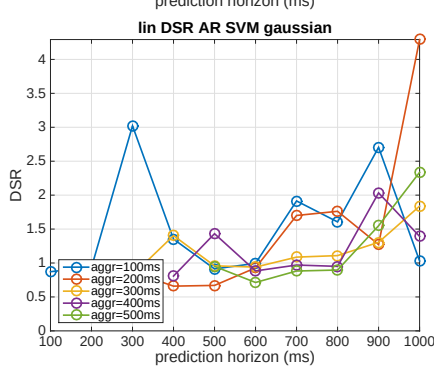
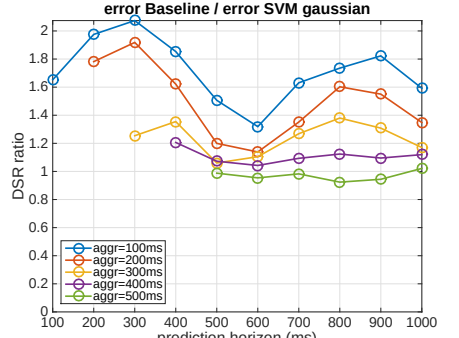
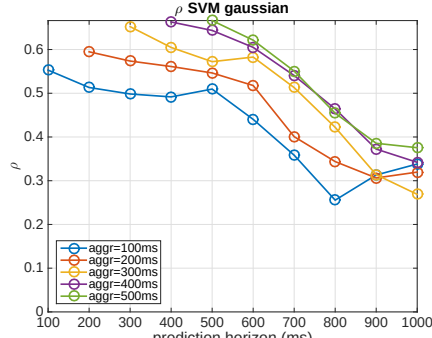
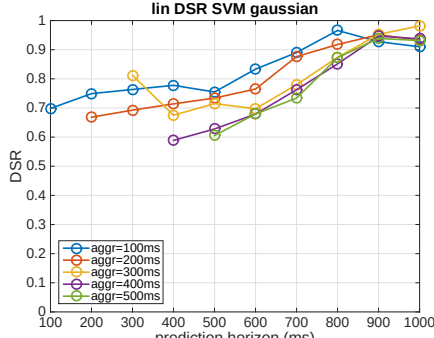
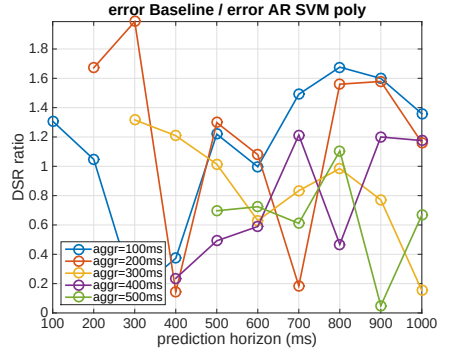
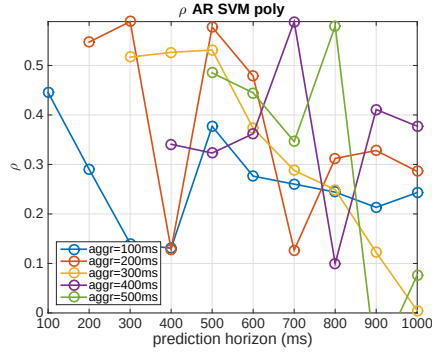
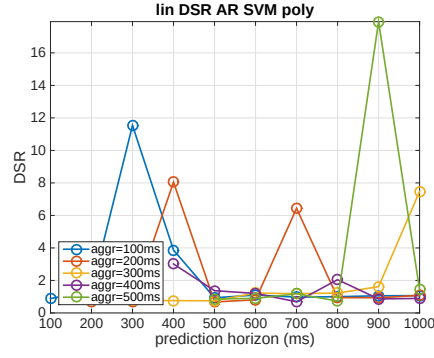
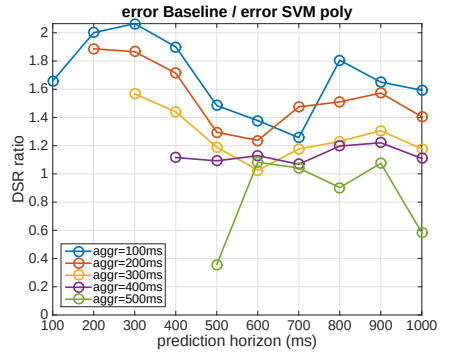
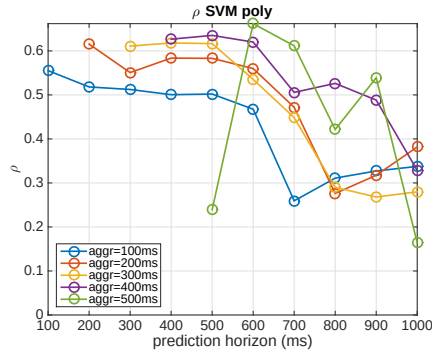
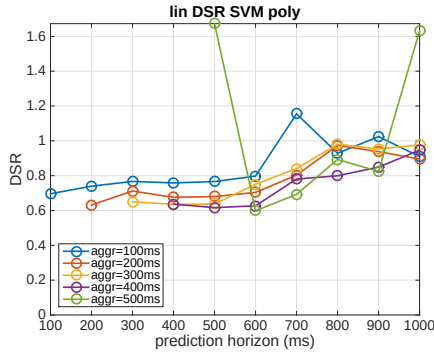


Résultats sur débit instantané agrégé

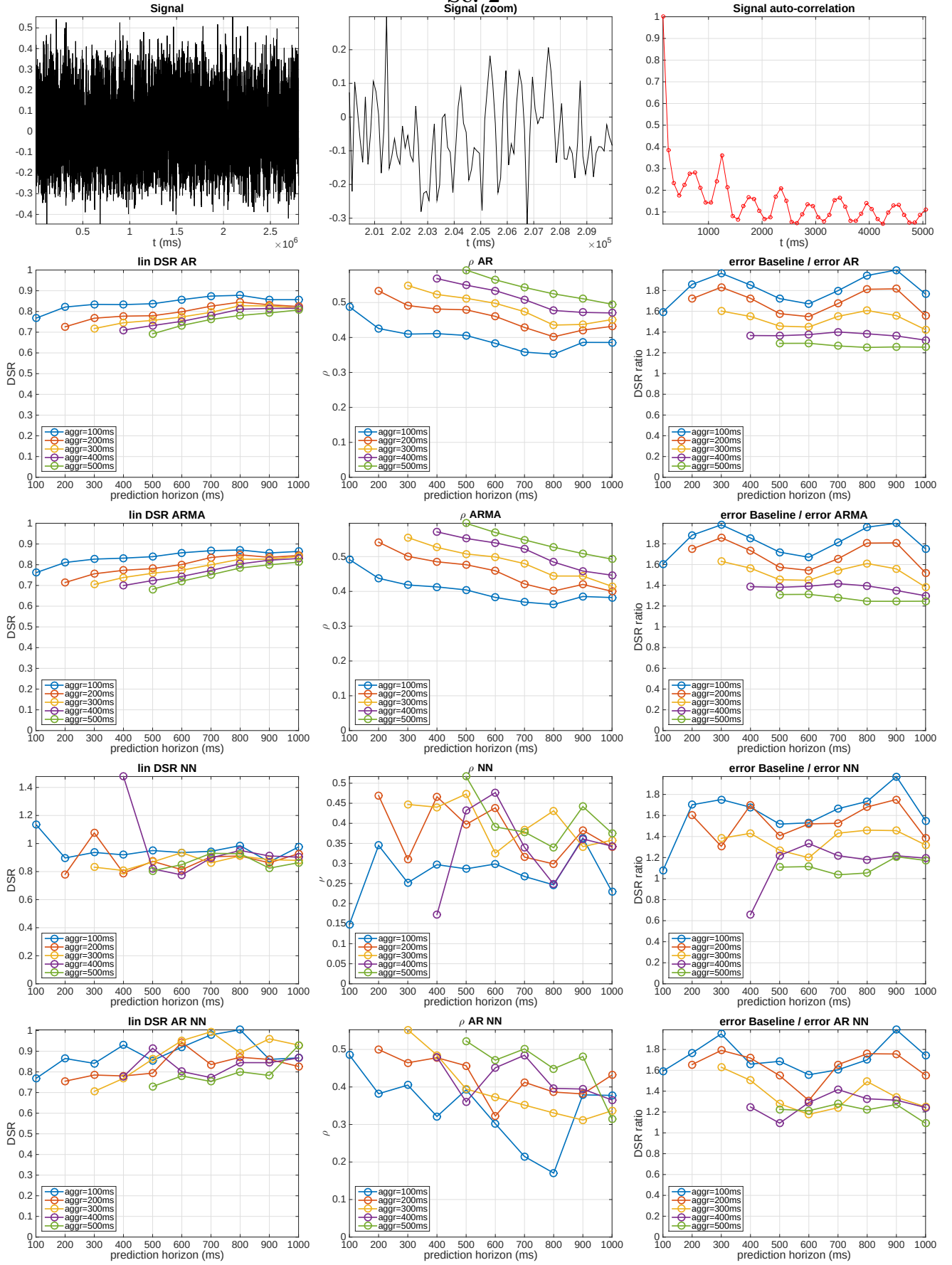
- Scénarios :
 - Sc. 1 Débit instantané échantillonné à 100 ms de flux agrégés simulé au niveau de la GW satellite (forward) pour un mélange de trafic pour 500 terminaux (28000 observations).
 - Sc. 2 Débit instantané échantillonné à 100 ms de flux agrégés simulé au niveau de la GW satellite (forward) pour un mélange de trafic pour 750 terminaux (28000 observations).
- Données :
 - agrégées sur des fenêtres glissantes de 100, 200, 300, 400, 500 ms
 - une 2e expérience est conduite sur la *dérivé* des signaux (i.e., prédiction du niveau d’augmentation/baisse du débit instantané, cf., p.6–9)
- Prédiction :
 - de la valeur agrégée
 - 100, 200, ..., 900, 1000 ms au delà de la dernière observation
(résultats affichés seulement pour prédictions au delà de la taille de la fenêtre d’agrégation)
- Protocol :
 - 500 échantillons d’entraînement
 - optimisation des paramètres via validation croisé sur les données d’entraînement (250 entraînement, 250 test)
- Methodes :
 - AR
 - ARMA
 - NN - réseau de neurones
 - SVM poly - SVM avec noyau polynomial
 - hybride AR NN - AR suivi de NN
 - hybride AR SVM poly - AR suivi de SVM poly
 - “baseline” : la dernière observation est utilisé comme prédiction
(SVM gaussien et hybride AR SVM gaussien n’ont toujours pas fini de tourner...)
- Quantification des performances :
 - “Distortion-to-Signal ratio” (DSR) - linear
$$\text{DSR} = \left(\|x - \hat{x}\|_2^2 / \|x\|_2^2 \right)$$
 - coefficient de correlation ρ entre signal et prédiction
 - rapport entre erreurs quadratiques $\|x - \hat{x}\|_2^2$ de la méthode en question et de la méthode “baseline”

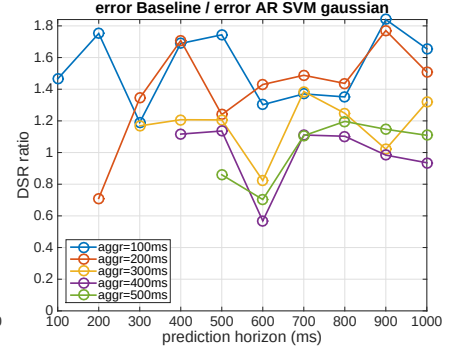
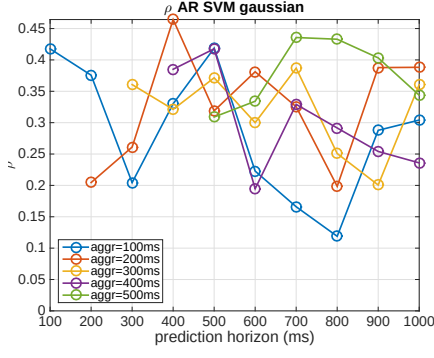
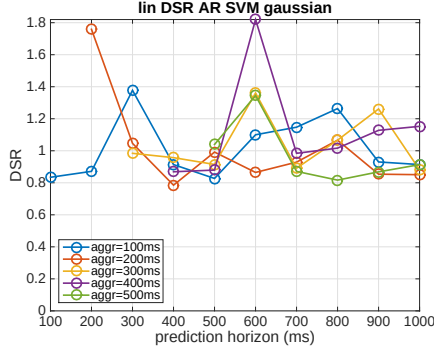
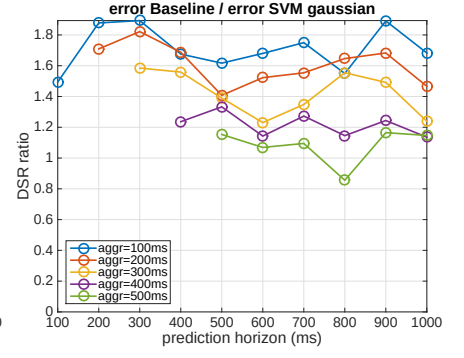
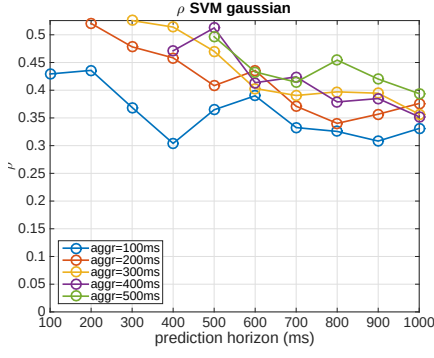
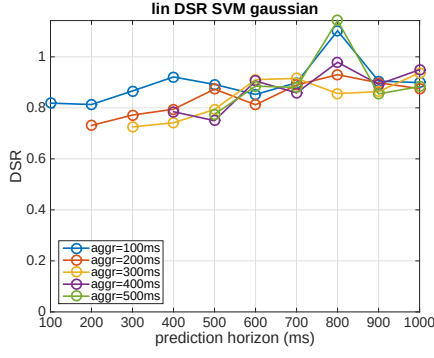
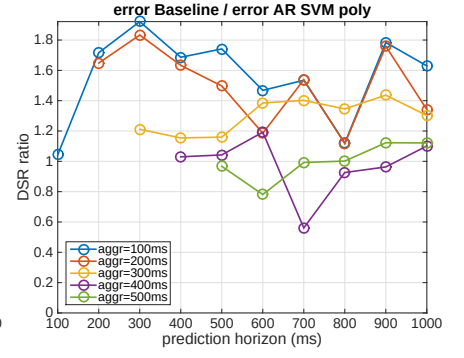
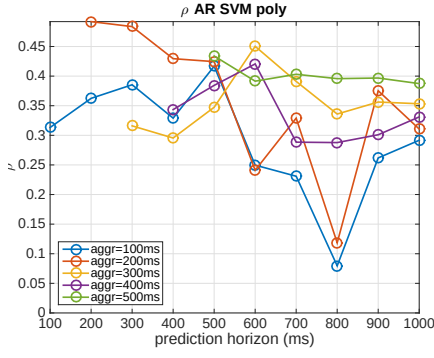
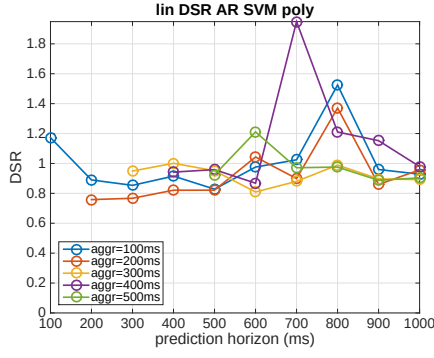
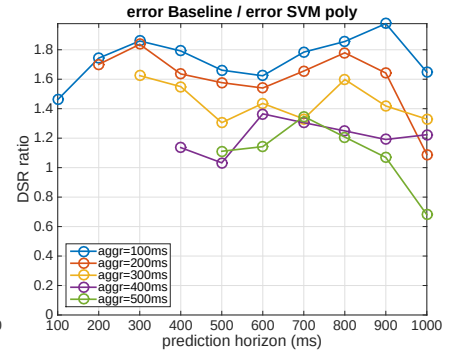
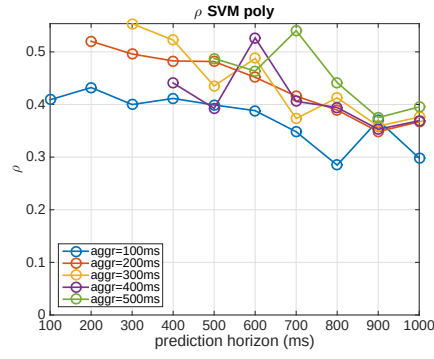
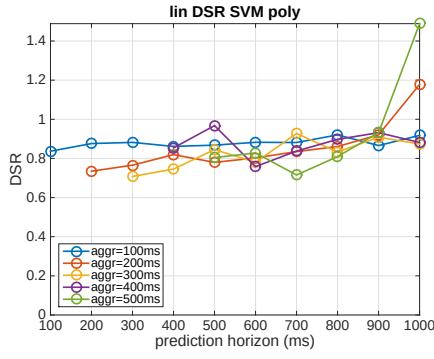
Sc. 1



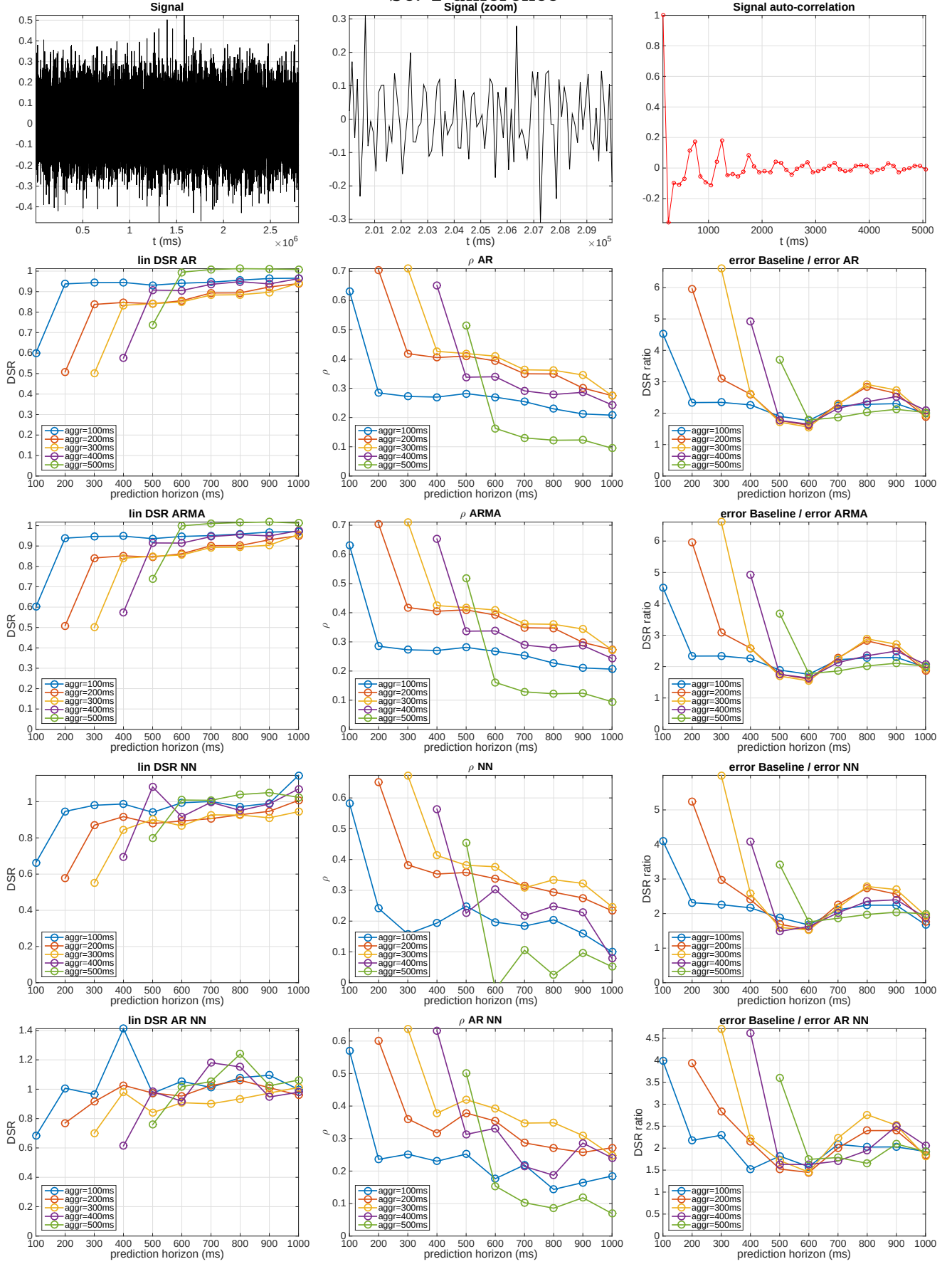


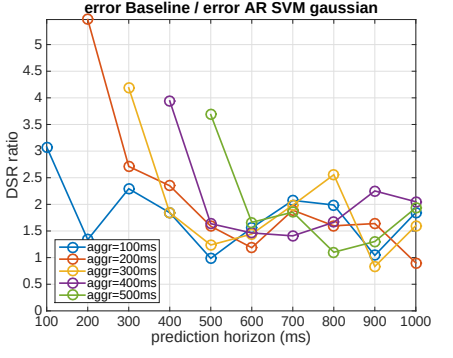
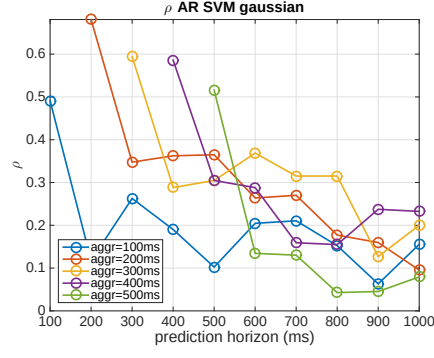
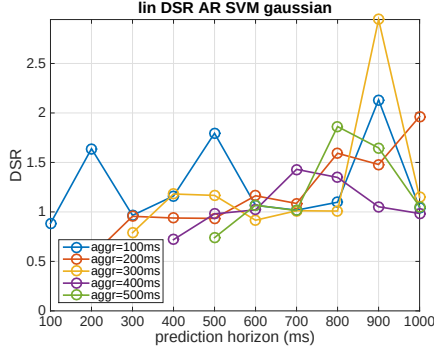
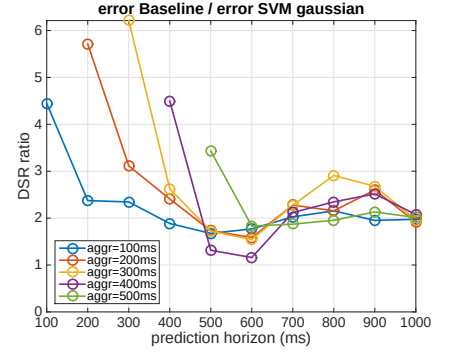
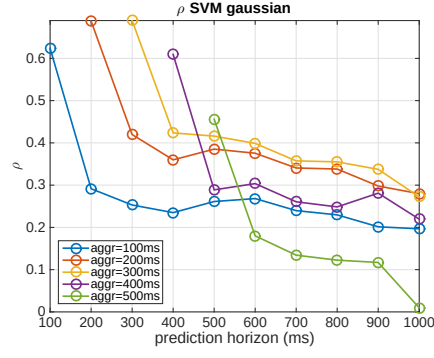
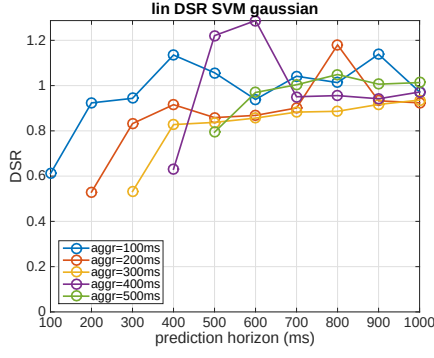
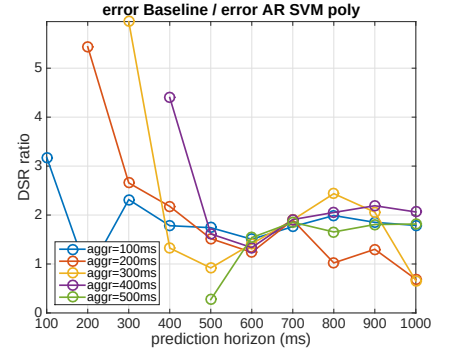
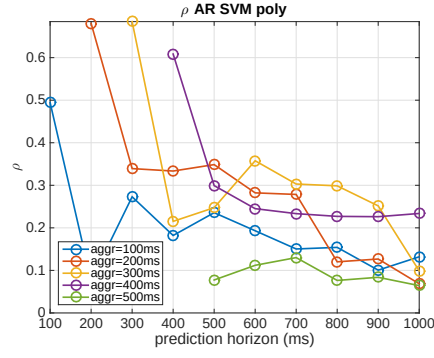
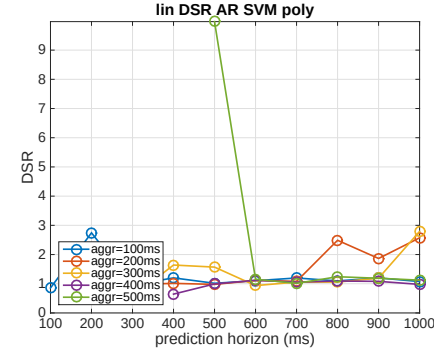
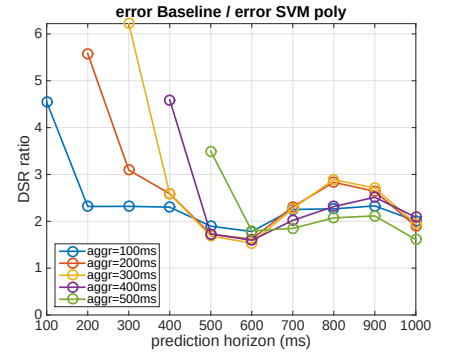
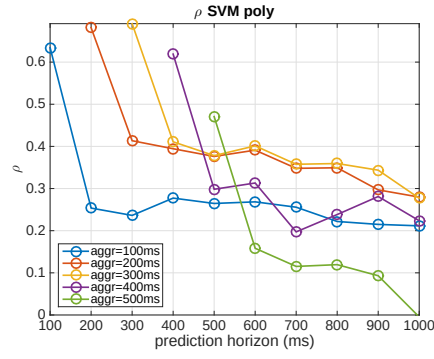
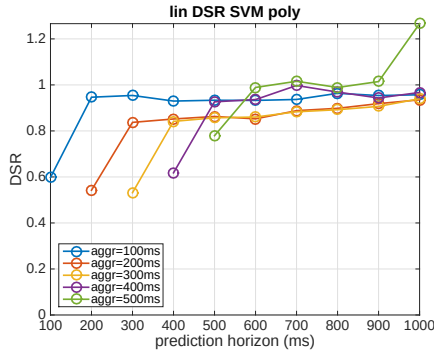
Sc. 2





Sc. 1 difference





Sc. 2 difference

